Bogleheads

Rebalancing

Rebalancing brings a portfolio that has deviated away from its target asset allocation back into line. The idea is that maintaining a consistent mix of asset classes (most commonly equities and fixed income) will help to keep risk at the desired level.

You do this by transferring funds from higher-performing classes to lower-performing classes. Although this may seem counterintuitive, rebalancing ensures that investors "Buy Low" and "Sell High".

Example

- You own two funds (Fund A and Fund B). The funds could be either mutual funds, index funds or exchange-traded funds (ETFs).
- You have a $10,000 portfolio and have selected an asset allocation such that each fund represents 50% of the total ($5,000 each)
- Over time, Fund A rises in value to $6000 while Fund B declines in value to $4,000 - you now have a 60/40 split instead of 50/50
- To maintain your asset allocation, you would rebalance your portfolio by taking $1,000 from Fund A and transferring it to Fund B
- After this, your asset allocation is once again 50/50

Different rebalancing approaches

There are several ways you can determine when it is time to rebalance:

- At a certain point in the calendar (for example, the beginning of the year, a specific day of the year, every other year, and so on). For example, you might systematically rebalance your portfolio once a year, on your birthday.
- When asset classes deviate from their target by a specific *absolute percentage*. For example, if your target asset allocation is 60% equities and 40% fixed income and your (absolute) rebalancing threshold is +/- 5%, you would rebalance your portfolio when your portfolio reaches (65% equities / 35% fixed income) or (55% equities / 45% fixed income).[note 1]
- When asset classes deviate from their target by a certain *relative percentage*. For example, if your target equities asset allocation is 60%, composed of 40% Total Stock Market and 20% Total International, and your (relative) rebalancing threshold is +/- 25%, you would rebalance International if it changes by more than +/- 5% (25% of 20%).[note 1]
- When asset classes deviate from their target by a certain dollar amount. For example, if you hold $6,000 in equities and $4,000 in fixed income and your rebalancing threshold is +/- $1,000, you would rebalance your portfolio when either of your holdings deviates from their target asset allocation of 60% equities / 40% fixed income by at least $1,000.
- When contributing to or withdrawing from your portfolio. For example, if your target allocation is 60% equities and 40% fixed income, you hold $7,000 of equities and $3,000 of fixed income, and you wish to contribute $1,000 to your portfolio, you would simply buy $1,000 worth of fixed income assets. This would bring you to an allocation of 64% equities / 36% fixed income. This approach minimizes transaction costs, effort, and taxes. This portfolio rebalancing calculator (https://optimalr

ebalancing.info) can help in cases where it is not obvious how much of a new contribution or withdrawal to allocate to each asset.

Other considerations

- Think about transaction costs when deciding whether to rebalance. Because transactions often have costs associated with them, many investors choose to wait for their portfolio to pass a significant threshold of deviation (whether defined by percentage or dollars) before rebalancing.
- Investors often find it difficult "selling winners" to "buy losers". To help remove emotions from the decision, you might choose a specific date to rebalance (for example, birthday, tax day, and so on). Investors who find the prospect daunting may want to consider Vanguard target retirement funds, which automatically rebalance as necessary to maintain a consistent asset allocation.
- The frequency with which you check your portfolio for deviations from its target has some influence on the returns gains (or losses) that rebalancing generates.[note 2]

Modified approaches

If you worry about rebalancing into a sustained bear market, one of these modifications to standard rebalancing methods might be useful:

- Keep a floor level of bonds, that is, once the total amount of bonds has dropped to a fixed amount, stop rebalancing into stocks.
- Only rebalance out of stocks, never into stocks.

Estimating changes in asset allocation

Normal market fluctuations do not frequently trigger rebalancing. For example, shifting the balance of a 50/50 portfolio by 1% requires a 4% change in the price of stocks relative to bonds.

All other portfolios are less sensitive, with 70/30 or 30/70 requiring a 5% change, 80/20 or 20/80 a 6% change, and 90/10 or 10/90 an 11% change. For example, a $10,000 60/40 portfolio will, after a 10% stock market drop, have $6,000 × 0.9 = $5,400 in stocks and $4,000 in bonds, for a stock allocation of $5,400 / ($5,400 + $4,000) = 57.45%, a 2.55% shift.

Notes

1. Larry Swedroe uses a "5/25 rule" which works like this:
   - **Absolute percentage of 5% for allocations 20% or greater:** When asset classes deviate from their target by an *absolute* percentage of 5%. For example, if your target asset allocation is 60% equities and 40% fixed income, you would rebalance your portfolio when your portfolio reaches (65% equities / 35% fixed income) or (55% equities / 45% fixed income).
   - **Relative percentage of 25% for allocations less than 20%:** When asset classes deviate from their target by a *relative* percentage of 25%. For example, if your target equities asset allocation is 60%, composed of 45% Total Stock Market and 15% Total International, you would rebalance Total International if it changes by more than +/- 3.75% (25% of 15%).

   See: Portfolio Rebalancing: The Whys And The Hows (https://seekingalpha.com/article/2130243-portfolio-rebalancing-the-whys-and-the-hows), Larry Swedroe. *Disable* cookies to view article on single page, no subscription required.

2. Research seems to indicate that checking weekly (while actually rebalancing very infrequently, for example by using large deviation thresholds) could provide the best rebalancing bonus. Conversely, rebalancing too frequently and using low thresholds would be detrimental. See: Opportunistic Rebalancing: A New Paradigm For Wealth Managers (https://pdfs.semanticscholar.org/57f8/766d83249e5be740c6cdacc8b30f8a5277cd.pdf), Gobind Daryanani.

Further reading

- Dynamic Strategies for Asset Allocation (https://web.archive.org/web/20220422055354/https://web.stanford.edu/class/msande348/papers/PeroldSharpe.pdf), Andre F. Perold, William F. Sharpe, Financial Analysts Journal, (January/February 1988):16-27. Archived from the original (https://web.stanford.edu/class/msande348/papers/PeroldSharpe.pdf) on April 22, 2022.
- Opportunistic Rebalancing: A New Paradigm For Wealth Managers (http://resource.fpanet.org/resource/09BBF2F9-D5B3-9B76-B02E27EB8731C337/daryanani.pdf), Gobind Daryanani, *Journal of Financial Planning*, (January 2008).

External links

- Bogleheads forum topic: "Bogleheads blog: a new rebalancing study" (https://www.bogleheads.org/forum/viewtopic.php?t=322344), siamond. August 8, 2020. A 3-part Bogleheads® blog series.
  - Part 1 (https://www.bogleheads.org/blog/2020/08/08/the-elusive-rebalancing-bonus-part-1/) explains and illustrates various rebalancing methods, then documents the data sources used in this study.
  - Part 2 (https://www.bogleheads.org/blog/2020/08/08/the-elusive-rebalancing-bonus-part-2/) analyzes if some of those methods provided a significant rebalancing bonus compared to an annual rebalancing model.
  - Part 3 (https://www.bogleheads.org/blog/2020/08/08/the-elusive-rebalancing-bonus-part-3/) explores miscellaneous topics (e.g. frequent rebalancing, less conventional asset allocations)
- Best Practices for Portfolio Rebalancing (https://web.archive.org/web/20230606014619/https://www.vanguard.ca/documents/best-practices-for-portfolio-rebalancing.pdf) by Vanguard Institutional Research, November 2015. Archived from the original (https://www.vanguard.ca/documents/best-practices-for-portfolio-rebalancing.pdf) on June 6, 2023.
- Getting back on track: A guide to smart rebalancing (https://web.archive.org/web/20220125212817/https://personal.vanguard.com/pdf/ISGGBOT.pdf) by Vanguard Research, April 2019. Archived from the original (https://personal.vanguard.com/pdf/ISGGBOT.pdf) on January 25, 2022.
- Portfolio Rebalancing: The Whys And The Hows (https://seekingalpha.com/article/2130243-portfolio-rebalancing-the-whys-and-the-hows) by Larry Swedroe, April 2014.
- Bogleheads forum topic: "How to use Vanguard Portfolio Watch for rebalancing" (https://www.bogleheads.org/forum/viewtopic.php?t=150267).
- Rebalancing (https://likelyso.com/rebalancing): a tutorial which discusses several methods of rebalancing, by Able to Pay (https://www.able2pay.com/about.html), discussed in this Bogleheads forum topic: "Rebalancing, another study" (https://www.bogleheads.org/forum/viewtopic.php?t=182926).

- Bogleheads forum topic: "Rebalancing: adaptive bands" (https://www.bogleheads.org/forum/viewtopic.php?t=186203), discussion about idiosyncrasies of absolute percentage and relative percentage approaches, suggesting a more 'adaptive' way to define rebalancing bands.

Spreadsheets

- Rebalancing Bands Spreadsheet (https://docs.google.com/spreadsheets/d/18HKaSlRFVlNhB1AG_npZO_aHh9mu_6QY5zKrKzZioKY/edit?usp=sharing) illustrating how to define 5/25 absolute/relative rebalancing bands across asset classes, on Google Docs from wiki contributor Hoppy08520, October 2014.
- Rebalancing spreadsheet (https://docs.google.com/spreadsheet/ccc?key=0AsVSNcUR-B6IdE4tZm0yVk9ObkxsZ2xvNmJpRTB0b0E), free download from Google Docs. Supplies three example approaches to rebalance a portfolio by allocation percentage, transfer amount, or final value; by forum member LadyGeek. To download, select File --> Download As --> Excel or OpenDocument.
- Boglehead rebalance.xls (https://drive.google.com/file/d/1bg9c7kxDEvHXDzEvCni7ethZ8uHwMJyd/view?usp=sharing) on Google Drive, by forum member LeeMKE. Rebalance your portfolio across multiple accounts. Includes space to keep notes.

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